

Design of a 5.4 kWp Grid-connected Residential Rooftop Solar PV system

(Project Brief)

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Location: Rajshahi, Bangladesh
System type: Grid-connected Solar system
Capacity: 5.4 kWp DC
Battery storage: 4.8 kWh nominal
Electrical supply: 1-phase LV (230V)

1.1 Project Overview

This project represents the conceptual design of a typical **5 kWp grid-connected residential rooftop solar PV system** for a residential building in Rajshahi, Bangladesh.

The purpose of the project is to demonstrate the basic engineering workflow involved in designing a small rooftop solar PV system, including site assessment, household load assessment, PV system sizing, module and inverter selection, basic electrical design, protection, earthing, rooftop layout and preparation of a bill of materials.

[Note that I'm not a professional nor an employer in Solar PV industry, so I don't have any prior experiences like site observation, case assessment etc., and also doesn't have real business data on my hand. But this doesn't stop me. I collected necessary data from Google and other AI platforms to stand this project and furnish my practice for such job positions]

The proposed system is designed as a **grid-connected system with battery storage**. Such model in my knowledge is more economical as it covers the daily household needs and also contributes to national grid, in optimal weather.

1.2 Project Objectives

- To assess the assumed residential electrical load.
- Determine an appropriate rooftop PV capacity.
- To select suitable PV modules.
- Develop a basic PV string configuration.
- Select a suitable single-phase hybrid inverter.
- To size a practical residential battery storage system.
- Estimate battery backup capability for selected essential loads.
- Determine basic DC and AC cable requirements.
- Identify basic protection and earthing arrangements.
- Prepare a rooftop PV layout.
- Prepare a basic electrical layout.
- To prepare a SLD & a basic BOM
- To demonstrate a practical residential hybrid solar PV design workflow.

1.3 Proposed System Data

This is the specifications I've considered for this project. However, the final electrical protection ratings, cable sizes and battery operating parameters will be determined during the design calculation stage.

Parameter	Proposed Value
PV System capacity	5.4 kWp
System class	Grid-connected hybrid
Battery capacity	4.8 kWh nominal, 48 V
Battery type	Lithium based with BMS
PV module	450W mono-crystalline

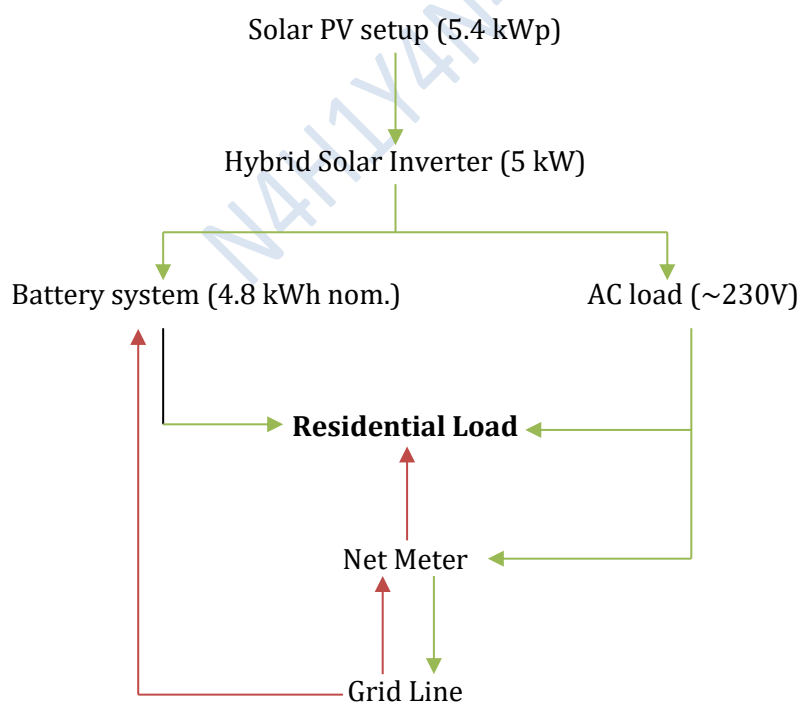
No. of panels	12 panels
PV configuration	6S + 6S
Inverter	GoodWe GW5000 ES
Inverter capacity	5 kW
Inverter Type	Dual MPPT block (~500V safety limit per block, 20A max per string)
Grid Line	1-phase 230V 50Hz
Mounting	Rooftop
Roof type	Flat RCC
Net Metering	Considered

The reason for choosing Mono-Crystalline panels over Poly-Crystalline type is because mono types are much more efficient (23%+ compared to 17% in poly), requires less space, has high heat tolerance that similar poly-crystalline panels. Only downside is, Mono-Crystalline panels are much expensive than Poly-Crystalline panels.

Each panel of 450 W and total of 12 panels gives us 5.4 KWp setup. The reason for selecting overall setup a bit higher than 5 kWp is because solar panels rarely meet the 100% capacity due to efficiency, environmental factors. So it's a good practice to have an upper margin. The design comprises 6 panels in series for MPPT-1 and another 6 panels in series for MPPT-2.

1.4 Basic Diagram

This is the skeleton of this project –



1.5 Site Description

The proposed system is assumed to be installed on the flat RCC rooftop of a residential building located in Rajshahi, Bangladesh. The assumed roof dimensions are approximately **10 m × 8 m**, giving a total roof area of approximately **80 m²**. The rooftop is assumed to have a generally south-facing usable area with low shading. A staircase room, water tank and parapet are considered when planning the PV array. All site dimensions and

conditions are assumed for portfolio project purposes. A real installation would require a physical site survey and verification of dimensions, shading, roof condition and structural suitability.

1.6 Design Assumptions

The preliminary design uses the following assumptions:

- The building has a single-phase LV electrical connection.
- The grid voltage is assumed to be 230 V, 50 Hz.
- The rooftop is assumed to be structurally suitable, subject to professional verification.
- The roof is assumed to have low shading.
- PV modules are assumed to face generally south.
- A hybrid inverter is used instead of a PV-only grid-tied inverter.
- Battery storage is included for energy shifting and selected backup loads.
- The battery is assumed to be a 48 V-class lithium battery with an integrated battery management system.
- A nominal battery capacity of approximately 4.8 kWh is assumed.
- High-power loads such as air conditioners are not considered essential backup loads.
- Final battery operating limits will follow the selected battery and inverter manufacturer's specifications.
- Net-metering/export operation is subject to applicable Bangladesh requirements and utility approval.

1.7 Preliminary Load Assessment

A typical residential load profile is considered here to estimate household electricity demand. The assumed loads include lighting, ceiling fans, refrigerator, television, computers, washing machine, water pump, air conditioning and miscellaneous loads.

The estimated daily energy demand will be calculated using assumed equipment ratings and operating hours. A separate **essential-load profile** will be developed for battery backup calculations. Essential loads will prioritize lighting, fans, refrigerator and selected small appliances while excluding high-power loads such as air conditioners.

1.8 Energy Storage Philosophy

The battery is included primarily for:

- Storing excess solar energy
- Shifting daytime solar energy to evening use
- Reducing unnecessary grid imports
- Providing backup power to selected essential loads during grid outages

The battery will not be treated as a means of powering the heavy household loads (like AC, Heater etc.) during an outage. For this project, a dedicated backup/essential-load concept will be used so that battery capacity is matched with realistic residential backup requirements.

1.9 Design Scope

The project scope includes:

- I. Residential load assessment.
- II. Rooftop site assessment.

- III. Solar resource assessment.
- IV. PV system sizing.
- V. PV module selection.
- VI. String configuration.
- VII. Hybrid inverter selection.
- VIII. Battery capacity selection.
- IX. Battery backup-load assessment.
- X. DC cable sizing.
- XI. AC cable sizing.
- XII. Basic DC and AC protection.
- XIII. Battery-side protection.
- XIV. Earthing and bonding concept.
- XV. Rooftop PV layout.
- XVI. Basic electrical layout.
- XVII. Single-line diagram.
- XVIII. Bill of materials.
- XIX. Final project report.

1.10 Exclusions

The following are outside the scope as this is not a real-world project:

- Structural certification.
- Actual utility approval.
- Actual net-metering application.
- Physical installation.
- Battery commissioning.
- Live electrical testing.
- Procurement.
- Detailed battery thermal design.
- Detailed lightning-risk assessment.
- Construction/stamped-for-approval drawings.
- Actual energy-generation measurements.
- Actual battery-cycle measurements.

1.11 Applicable References

The design is considered by applicable Bangladesh standard requirements and relevant IEC-based requirements for photovoltaic systems, grid-connected inverters and electrical installations.

For Residential load calculation (Diversity load), I followed BNBC classifications. The current SREDA equipment listing includes approved single-phase grid-connected/hybrid inverter models and identifies applicable standards such as IEC/BDS IEC 61727, IEC/BDS IEC 62116 and IEC/BDS IEC 62109 for listed inverter models. However, final equipment selection for an actual installation must be checked against the latest applicable SREDA approved-equipment requirements.

1.12 Expected Deliverables

The completed portfolio project will contain:

- Project Brief
- Site Assessment
- PV Design Calculations

- Electrical Design Calculations
- Bill of Materials
- Rooftop PV Layout
- Basic Electrical Layout & SLD

1.13 Design Philosophy

This is usually what professionals follow to complete a Solar PV installation project:

Site Assessment → Load Assessment → PV Sizing → Module Selection → String Design → Hybrid Inverter Selection → Battery Sizing → Cable & Protection Design → Earthing → CAD Drawings → BOM → Final Report

However, I don't have the expertise on selecting inverter, specific battery packs or cable selection. So, I will consider these parts as general assumption and focus more on CAD drawing, load and power generation calculations, BOM and final report.

1.14. Conclusion

This project demonstrates the basic engineering process involved in developing a residential hybrid solar PV system.

The proposed system combines a **5.4 kWp PV array, 5 kW single-phase hybrid inverter and approximately 4.8 kWh battery storage**. The system is intended to demonstrate solar generation, battery energy storage, household energy supply, selected backup operation and grid interaction. All calculations and equipment selections will be verified during the detailed design stage.